AP CALCULUS AB

Unit 1 • Lesson 1.12 practice

Lesson 1.12 • Confirming Continuity over an Interval

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

On which interval is $f(x) = \sqrt{x-3}$ continuous?

(A) $[3, \infty)$
(B) $(3, \infty)$
(C) $(-\infty, 3]$
(D) All real numbers

Question 2

NO CALCULATOR

Where is $f(x) = \dfrac{x+1}{x^2-4}$ continuous?

(A) All real numbers
(B) All real numbers except ± 2
(C) All real numbers except -1
(D) All real numbers except 2

Question 3

NO CALCULATOR

On which interval is $f(x) = \ln(x+5)$ continuous?

(A) $(0, \infty)$
(B) $[-5, \infty)$
(C) $(-5, \infty)$
(D) $(5, \infty)$

Question 4

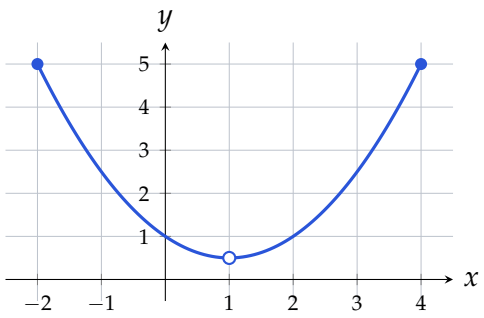
NO CALCULATOR

On which interval is $\tan x$ continuous?

(A) $[0, \pi]$
(B) $(0, \pi)$
(C) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Question 5 (graph) NO CALCULATOR

The graph of f on $[-2,4]$ is shown; it has a hole at $x = 1$. On which of the following intervals is f continuous?



(A) $[-2,0]$
(B) $[0,2]$
(C) $[-2,4]$
(D) $[1,3]$

Question 6 (Free response) NO CALCULATOR

Let $h(x) = \dfrac{\ln x}{x^2 - 9}$.

(a) Find the domain of h, and explain why h is continuous on its domain.
(b) Is h continuous on the closed interval $[1,2]$? On $[2,4]$? Explain.